

16 D

Intensity $I = \frac{P_{\text{source}}}{\text{Area}} = \frac{P_{\text{source}}}{4\pi x^2}$ (As the source is a point source, the Area through which energy pass through is the surface area of a sphere with radius x.)

As P_{source} is constant, $I \propto \frac{1}{x^2}$

$$\frac{I_1}{I_2} = \left(\frac{x_2}{x_1}\right)^2 \Rightarrow \frac{I}{I_2} = \left(\frac{0.5x}{x}\right)^2 \Rightarrow I_2 = 4I$$

As $I = \frac{P_{\text{received}}}{\text{Area}_{\text{reciever}}}$

$$P = IC$$

$$P_{\text{received } 2} = I_2 \times \text{Area}_{\text{reciever } 2} = 4I \times 2C = 8P$$

17 D Using Rayleigh's Criterion